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PyGraphon: a Python package for graphon estimation

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Summary

describing the high-level functionality and purpose of the software for a diverse, non-specialist audience

Pygraphon is a Python library designed to estimate graphons, providing an intuitive interface for researchers and practitioners to model and analyze graph structures efficiently. Graphons are probability distributions on the space of graphs, allowing users to infer underlying network characteristics from observed data. This library caters to a broad audience, enabling non-specialists to engage with graphon estimation techniques and apply them in various fields, including social networks, neuroscience, and machine learning.

Statement of need

clearly illustrates the research purpose of the software and places it in the context of related work.

Graphon estimation stands as a crucial tool in the analysis of large-scale networks, serving as a fundamental step in understanding complex systems. Pygraphon addresses the need for accessible, user-friendly tools that enable researchers across domains to estimate graphons accurately. By providing a comprehensive set of functionalities for graphon modeling, estimation, and visualization, this library contributes to the advancement of network science and facilitates insights into the structure and dynamics of complex networks.

Similar software

A list of key references, including to other software addressing related needs. Note that the references should include full names of venues, e.g., journals and conferences, not abbreviations only understood in the context of a specific discipline.

- R package `graphon` (You, 2017)
- R package `nethist` *(development stage)* (Song & Olhede, 2023)
- `Julia` package `NetworkHistogram.jl` (Dufour & Grainger, 2023)

We list the different methods implemented in `Pygraphon`, `graphon`, and `NetworkHistogram.jl` latter in Table 2.

Mention (if applicable) a representative set of past or ongoing research projects using the software and recent scholarly publications enabled by it.

Recent Scholarly Publications Enabled by `Pygraphon`:

- Verdeyme & Olhede (2024)

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Functionality and implementation

For more details, please refer to the documentation.

$$A_ij|\xi_i, \xi_j \sim \text{Bernoulli}(W(\xi_i, \xi_j))$$

Methods	class names	Reference
Matrix completion	`Completion`	(Keshavan et al., 2010)
Largest Gap algorithm	`LG`	(Channarond et al., 2012)
Sort and Smooth	`SAS`	(Chan & Airoldi, 2014)
Network Histogram	`HistogramEstimator`	(Olhede & Wolfe, 2014)
Neighborhood smoothing	`NBDsmooth`	(Zhang et al., 2017)
USVT	`USVT`	(Chatterjee, 2015)
Shape Histogram	`SmoothNetHist`	(Verdeyme & Olhede, 2024)

Table 1: Comparison of different methods implemented in `Pygraphon` with their corresponding class names and references.

Comparison with existing implementations

`nethist` and `NetworkHistogram.jl` only implement the network histogram method.

	`Pygraphon`	`graphon`
Matrix completion	Yes	Yes
Largest Gap	Yes	Yes
SBA[1]		Yes
Sort and Smooth	Yes	Yes
Network Histogram	Yes	Yes
Neighborhood smoothing	Yes	Yes
USVT	Yes	Yes
Shape Histogram	Yes	

Table 2: Comparison of different methods implemented in `Pygraphon`, `graphon`, and `NetworkHistogram.jl` with their availability indicated by "Yes".

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[1] the SBA algorithm (Airoldi et al., 2013) is not implemented in `Pygraphon` yet as it is intended to be used with multiple graphs and not a single observed graph. We plan to implement it in the future.

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